The DesignMind Toolkit

A toolkit for evidence-based, cognitively-informed and human-centred architectural design

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Integrating empirical evidence into architectural design has the potential to create more humane, legible, and inclusive spaces. Yet, the fast-paced nature of the design process, limited proficiency in scientific terminology, and a natural reliance on design intuition limit architects' ability to effectively apply empirical findings to inform their design decisions. To address this gap, this paper presents DesignMind, a toolkit for evidence-based, cognitively-informed, and human-centred architectural design. DesignMind has four complementary modules: (1) evidence-based design flashcards, (2) topological and geometric analysis, (3) agent-based simulation, and (4) cognitive walkthrough in virtual reality. The effectiveness of DesignMind was evaluated in an academic context where architecture and STEM students used it to analyse and design healthcare and workplace layouts. Initial results show that DesignMind is effective in integrating evidence-based design into the architectural design process and providing students with tools to test their designs for human-centred design criteria such as wayfinding, social communication potential and nursing efficiency. These preliminary results underscore the challenges and opportunities of DesignMind to fuel parallel efforts in evidence-based design, calling for a critical discussion around the notion of evidence in architecture and its interplay with design intuition.

Keywords: *Evidence-based Design, Design Education, Human-Centred Architecture, Design Toolkit, Design Cognition*

INTRODUCTION

Intuition can be a powerful tool for problem-solving, and prior research has confirmed that experienced architects use abductive or appositional thinking that is closely linked to intuition to resolve ill-defined design problems as acknowledged by Cross (1982, p. 221-227) and Kalay (1999, p. 395-409).

However, designing built environments that balance environmental, human-centered, and economic goals, along with the increasing empirical evidence linking physical design and human-related outcomes, calls for Evidence-Based-Design (EBD) approaches which emphasize using the best available evidence from systematic research in making decisions as defined by Ulrich et al. (2008, p. 61-125). Yet, despite its potential value for architecture, EBD is often contrasted with intuition, or the 'designerly ways of knowing' as coined by Cross (1982, p. 221-227), and is criticized for being non-integrative and possibly even limiting design creativity.

Acknowledging the tension between these two approaches as emphasized by Archer (1979, p. 17-20) and Nadler (1967, p. B-642), the primary objective of this paper is to explore the integration of EBD and design intuition in the architectural design process.

Module	*Prior Knowledge*	*Source of Evidence*	*Human Unit of Analysis*	*Method*	*Individual vs. Co-Design*	*Dev. Environment*	*Ease of Use*
1	Low	Empirical Literature	Diverse	Flashcards	Individual	Physical Deck	Easy
2	Moderate	Empirical Literature	Spatial	Spatial Metrics	Individual	Grasshopper for Rhino	Medium
3	High	Empirical literature	Social	Behaviour Model	Co-Design	Unity3D, C# Scripts	Hard
4	Moderate	HCI	Individual	Cognitive Walkthrough	Individual	Unity3D	Medium

Table 1 Comparison of *DesignMind* modules with respect to prior knowledge required, unit of analysis, individual vs. co-design, method of linking evidence to design, development environment, and ease of use.

We seek to bridge the gap between intuition and evidence and investigate how they can be effectively combined to inform architectural design decision and outcomes.

Towards this end, we present *DesignMind*, a toolkit to reinforce design intuition with empirical evidence and human-centred analytics grounded in recent research on spatial cognition showcased in Gath-Morad et al. (2022, p. 104131; 2021, p. 18980; 2020, p. 13-23; 2020, p. 1-8), Grübel et al. (2022, p. 101640; 2021), and Schaumann et al. (2016).

Our toolkit is targeted at architecture students as well as practising architects and designers who seek to leverage evidence-based approaches to augment their intuitive design thinking. The paper outlines the modules composing *DesignMind*, focusing on the methods that underlie it and including examples of how the toolkit was used and evaluated in an academic context.

Finally, we conclude the paper by highlighting the challenges and opportunities of *DesignMind* to fuel parallel efforts in evidence-based design. We call for a critical discussion around the notion of *evidence* in architecture and its interplay with design *intuition*.

METHODS

DesignMind consists of four main modules (Figure 1); (1) Evidence-based design flashcards, which offer access to a curated collection of evidence-based design literature organized by design criteria such as wayfinding, privacy, productivity, and collaboration; (2) Topological and Geometric Analysis, providing quantitative analysis of circulation and visually perceived space shape; (3) Agent-based Simulation, which simulates occupants' movement to identify interactions and bottlenecks; and (4) Cognitive Walkthrough in Virtual Reality, allowing for recording, visualization, and analysis of virtual task-based walkthroughs from different perspectives.

Each module targets a different aspect of the design process and uses different methods and tools to link design evidence to design decisions. Table 1 summarises the four modules and compares them with respect to prior knowledge required, unit of analysis, individual versus co-design, method of linking evidence to design, development environment, and ease of use.

The first module, designed for designers with low prior knowledge, uses flashcards to introduce architectural terms and concepts. The second module, for designers with medium prior knowledge, focuses on spatial metrics and uses Grasshopper for Rhino to provide geometrical and topological analysis of design layouts. The third module, for designers with high prior knowledge, employs behavioural modelling to simulate human behaviour and uses Unity3D and C# scripts to implement the agent-based simulations. Finally, the fourth module, designed for individual learners with

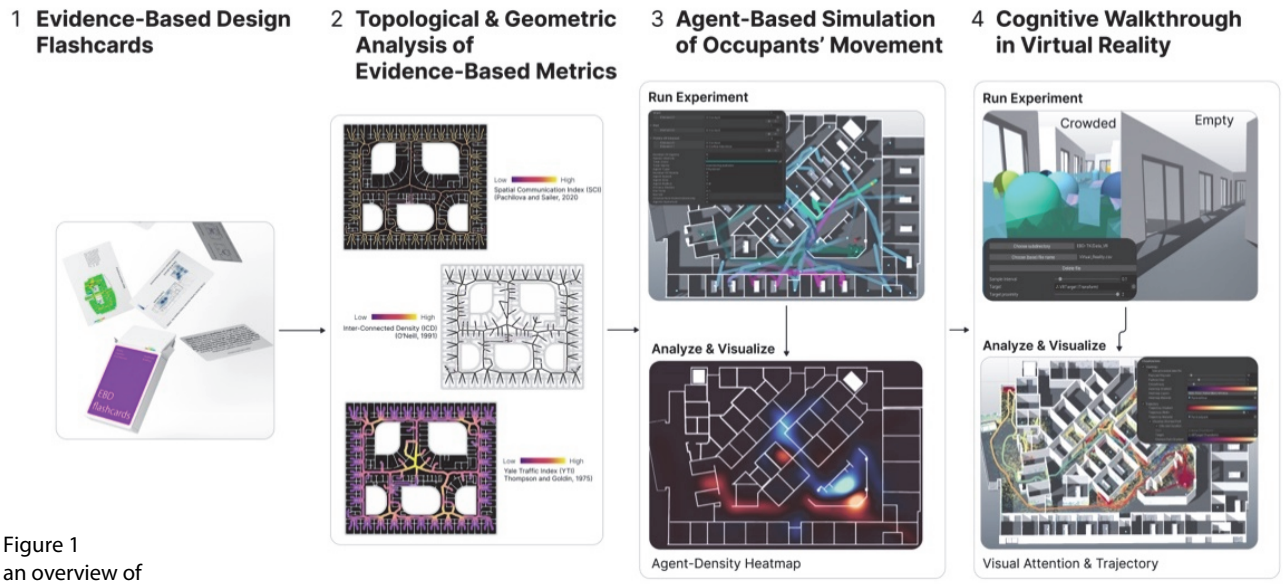

Figure 1
an overview of the four modules composing DesignMind
Source: authors.

medium prior knowledge, uses cognitive walkthroughs, a method which is often used in Human Computer Interaction (HCI) (Krukar et al., 2006, p. 17-35) to identify potential design issues and uses Unity3D to create a virtual environment for performing the simulations.

Modules also differ from one another in their unit of analysis, with some targeting individual designers and others focusing on group collaboration or co-design. The development environments for each module also vary, with some using Unity and C# scripts and others using Grasshopper for Rhino. Finally, the modules differ in terms of their ease of use, with some being relatively easy to use (e.g., flashcards) and others requiring more advanced technical skills (e.g., agent-based simulations). Overall, DesignMind provides a range of tools and methods that can support designers with different levels of prior knowledge and different learning styles in the architectural design process. This comparison could help appropriate module(s) for different design context, pedagogical goals, and levels of design experience.

(1) Evidence-based design flashcards

The *Evidence-based Design Flashcards* module is designed to provide designers with access to a curated collection of evidence-based design literature organized by design criteria, including wayfinding, privacy, productivity, and collaboration. This module is greatly inspired by the previous research that developed a deck of architectural cognition cards to introduce spatial cognition research and user-centred thinking into the design process as emphasized by Mavros et al. (2022, p. 120-137).

The flashcards developed as part of *DesignMind* consist of a physical deck and a virtual collection of cards, making them easy to use and accessible individually or in a co-design context, without the need for specialized software or technical knowledge. The cards focus on EBD in healthcare

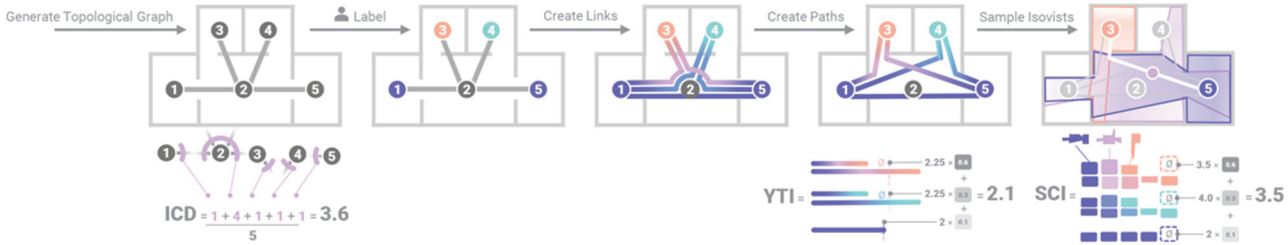

Figure 2: Computational process of calculating the three spatial measures (ICD, YTI, SCI) included in the geometrical and topological analysis module from left to right).
Source: authors.

environments, whereby each card reviews one paper (by extracting take away results, methods, and figures.

By providing a quick reference to relevant research on specific design criteria, this module aims to inform the design process with evidence-based principles and promote best practices in architectural design. The low prior knowledge requirement and easy-to-use format of the flashcards make them a valuable resource for individual designers seeking to incorporate evidence-based design principles into their work.

(2) Topological and geometric analysis of evidence-based metrics

The *Topological and Geometric Analysis* module was implemented in Grasshopper for Rhino by Robert McNeel & Associates (2023) and takes as input a floorplan of a layout to generate several scores of EBD metrics that have been empirically linked with human-centred outcomes. These include InterConnected density (ICD) as proposed by O'Neill (1991, p. 553-574), The Yale Traffic Index (YTI) introduced by Thompson and Goldin (1975, p. 282-295), and The Spaces for Communication Index (SCI) developed by Pachilova and Sailer (2020, p. 186-202).

ICD is a measure of layout complexity that quantifies the average number of neighbouring nodes for each node in the topological graph. Low ICD values indicate low complexity, while high ICD values correspond to high complexity. YTI measures layout efficiency by determining the expected walking distance of trips performed by nurses based on the layout and its spatial configuration. Low YTI values correspond to high efficiency, while high YTI values indicate inefficient layouts. SCI captures the potential of spaces to facilitate face-to-face interactions, which positively impact patient outcomes. SCI corresponds to the average visibility along the most frequented links in a layout, weighted by the relative frequency of these links and normalized by the number of patient beds.

We computed these indicators based on the underlying topological graph, an abstract representation of the configuration of spaces in the layout. In this graph, spatial partitions, e.g., rooms, correspond to nodes, and nodes are connected if the corresponding partitions are directly adjacent. We resorted to a formal description of said computation based on the medial axis transform given by authors in (Carranza and Koch, 2013, p. 54-1), and to our knowledge, provide the first fully Grasshopper-based implementation. Students can assign a room type to each node, e.g., nurse station, in a dedicated interface that shows the graph and node labels superimposed on the floor plan. With that, the graph captures not only topological but also semantic information.

Upon each change in the layout, the graph is recomputed and thus allows for interactive exploration of the relationship between the floorplan and the topological graph and the following derived metrics. As for SCI, it is calculated by spatializing the most frequented links using the

annotated topological graph and a geometric computation of isovists (a viewshed polygon from a single vantage point) along these paths, subsequently averaging their areas. This process is presented in Figure 2.

3) Agent-based simulation of occupants' movement

The *Agent-Based Simulation* module requires input of a 3D building model with basic configuration elements (e.g., walls, openings) required to compute movement. The simulation environment consists of three main components, which were all implemented as standalone C# scripts and endowed with a user-friendly UI that exposed parameters to customize the simulation. The task component defines the properties of a single behaviour that the agents exhibit. In the task script, designers can describe the start stage in which agents would appear, intermediate sub-stages in which agents visit points of interest (POI), and an end stage where agents disappear from the simulation interface.

For each of these, designers can indicate multiple locations and whether agents would visit them in a predefined or randomized order. Designers can simulate the dynamics of different occupancy groups by specifying an agent type, optical parameters such as an agent colour, the number of agents associated with this behaviour, the delay with which the agents appeared in the simulation, and various behavioural parameters, such as walking speed, the amount of time spent at POIs, the degree of privacy the agents require (simulated by enforcing a collision-free zone around the agents). Agents followed a shortest-path heuristic to find their way between goals.

However, despite this simple pathfinding method, the randomization option facilitated the occurrence of rich emergent phenomena in the simulation. Agents follow a shortest-path heuristic to find their way between goals. The engine component allows students to register behaviours previously defined in the task component and set global simulation parameters. The engine script saves snapshots of the simulation state, i.e. the locations of all agents deployed in the simulation, at regular intervals.

Lastly, the visualizer component shows and analyses the data generated in previous simulations. The tool allows users to filter the data to be processed by role and agent type for fine-grained control over the analysis outcomes. By performing kernel density estimation, the component visualizes the agent distribution during the simulation as 2D or 3D heatmaps, allowing for, e.g., immediate identification of "hotspots" that can indicate bottlenecks in a layout. The tool also allows for comparison between distributions of agent groups by computing comparison heatmaps, to gain a qualitative understanding of how different behaviours might result in differences between agent distributions. The visualizer component also provides a rich output in terms of simulation statistics, including the average duration of completing tasks; average walking distance of agents performing tasks; average number of collisions within and between different agent groups. The output analysis is in the form of a CSV file shows aggregated statistics on different occupancy groups.

(4) A cognitive walkthrough in VR

The *Cognitive Walkthrough* module allows designers to experience their layouts from a first-person perspective. The input required to run the walkthrough is a 3D building model, a persona (representing the characteristics of a specific user), a task description, a starting location, and a target location in the layout. The cognitive walkthrough method is a technique commonly used in Human-Computer Interaction (HCI) that involves simulating the experience of different users as they perform tasks in a virtual or physical environment. This method allows designers to anticipate and identify potential issues and challenges that different users may encounter and adjust the design accordingly.

Upon encountering the target location, the walkthrough would automatically terminate.

Designers control their characters using the WASD and mouse input combination commonly found in many contemporary computer games. During the simulation, the position and view angle of the character is continuously saved to a CSV file. The output analysis resulting from the walkthrough is two visualizations representing the persona's movement and visual attention. In the first one, character trajectories, as well as a hypothetical shortest path trajectory between the present start- and target position, are displayed in the simulation environment. The second visualization shows the visual attention patterns spatialized in the layout. To that end, we projected particles from the recorded character positions along the direction of the corresponding viewing angles. The particles are coloured based on a kernel density estimate, gauging the intensity of visual attention directed at specific objects in the environment. In addition to the visualizations, the tool provides a set of informative measures about the characteristics of the walkthrough, including: the distance of the walkthrough; the surplus distance and distance ratio compared to the shortest path, providing a measure of efficiency; the duration of the walkthrough, indicating a level of directedness.

TESTING *DESIGNMIND* IN ARCHITECTURAL DESIGN PEDAGOGY

Course Description

The *DesignMind* toolkit was used in a semester-long course that targeted both undergraduate and graduate students in architecture and STEM fields. The course focused on analysing the human-centred aspects of healthcare and workplace environments. Students were tasked with selecting an existing pair of hospital or workplace typologies and analysing them using the *DesignMind* toolkit. Based on their analysis, students suggested design variations to test specific design hypotheses and then reapplied the toolkit to analyse their own designs. An overview of the students works is available in Figure 3, showing a cross comparison of spatial metrics (using module 2 of Design Mind) across hospital typologies.

Evaluation Approach

A teaching evaluation survey was conducted at the end of Fall 2021 semester. The survey was designed to provide a comprehensive evaluation of the course and included open-ended questions to gather qualitative data from the 13 students who participated in the course. As part of this short paper, we focus on thematic analysis of responses to two open-ended questions (1) how has your participation in the course affected your perspective of architecture, and (2) which aspects of human-centred design would you like to learn more about? The responses to these questions were analysed using content analysis (White et al., 2006) to identify the prominent themes and sentiment expressed by the students.

Preliminary Results

The responses of the students were analysed based on the themes that emerged from the open-ended questions. Several students expressed that the course provided them with new tools and methods to approach the architectural design process. One student noted that the course "has broadened my range of instruments with which to approach a design process, which I will try to implement in future designs."

Another student appreciated that the course highlighted "the importance of considering how people will use a space and feel inside it." This sentiment was echoed by several other students who emphasised the need to integrate occupant behaviour into the design process. As one student noted, "how people move in space is never how we think they will and doing the walkthroughs gave us a good picture of this." Another student found that the agent-based simulation was the most relevant tool for analysing occupant behaviour, stating that "one thing, which could be very well measured with the agent-based analysis, were the collisions of

Figure 3:
An overview of students' typological analysis using the topological and geometric analysis module of Design Mind. Floorplans describe existing hospitals compared for SCI, YTI and ICD. Authors: Student works (Julie Bovier, Laurianne Chassot, Florian Hofmann, Anju Kato, Roc-Andrea Rüegg, Florian Hofmann, Thierry Dvoracek, Ogunsola Oluwasegun), EBD class, ETH Zurich. Source: authors.

occupants and it was interesting what influence that had on other aspects of the behaviour."

Several challenges were also identified using the thematic content analysis. The most prominent challenge was related to the time and effort required for the course. Some students felt that the course was too time-consuming, and that the workload should be adjusted to reflect the number of credits assigned to the course. For example, one student mentioned, "I would think about giving 5 credits next semester as the time effort was much higher than other 3 credit courses." Another student expressed, "While the course opened up a whole new world of possibilities in terms of using simulation tools to inform design decisions, I found myself struggling with interpreting the quantitative results. I realised that I lacked the necessary scientific knowledge to fully understand the simulations and make informed design choices based on them. I think there is a need for more education and training in this area, especially for those of us who come from a more artistic or design-focused background." Additionally, some students felt that the course was mainly intended for graduate students and that it should be better advertised to reflect this.

With respect to intuition versus evidence, one student noted, "The tools we used were very informative and helped me think more critically about my design decisions." However, there were also some students who expressed concerns about relying too heavily on evidence-driven design and not enough on intuition. One student remarked, "I appreciate the usefulness of the tools but worry that they can sometimes be limiting and prevent me from following my instincts as a designer." Overall, it seems that the students found value in using both evidence-based tools and intuition in their design process, and that a balance between the two is necessary for effective design.

Finally, the thematic analysis highlighted the role of visualisations provided by *DesignMind* to facilitate communication. As one student stated, "It was eye-opening to see how small changes in the design could have a big impact on occupant behaviour. The visualisation tools helped me to communicate these differences to others on the team, and I feel much more confident in presenting design decisions with quantitative evidence to back them up." This sentiment reflects the role of *DesignMind* as a tool that supports communication and reflection on the design process. By providing visualisations and quantitative evidence, it helps to bridge the gap between intuition and evidence-based decision making, enabling more effective and efficient design solutions.

Discussion and Future Work

In conclusion, *DesignMind* provides a valuable toolkit for evidence-based design that can be used by designers with different levels of knowledge and for different purposes. However, as highlighted by the student feedback, there are also challenges that need to be addressed, such as the need for more education and training in interpreting and using quantitative simulation results. It is important to strike a balance between evidence-based design and intuition (Cross, 1982, p. 221-227), and to have a critical discussion around the notion of evidence in architecture. Additionally, the fact that the toolkit is open-source provides opportunities for further development and customization to meet the needs of specific contexts and user groups. Future work should focus on testing *DesignMind* in practice and gathering feedback from designers who have used it in their work. We also encourage educators in architecture and related fields to test the toolkit with their students, to enhance their knowledge and skills in evidence-based design.

DATA AVAILABILITY

The *DesignMind* toolkit is available here:
https://doi.org/10.5281/zenodo.7791794

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